2D plotting commands

`figure:`	Open a new figure window.
`Subplot:`	Multiple axes in one figure.
`semilogx`, `semilogy`, `loglog:`	Logarithmic axis scaling.
`axis`, `xlim`, `ylim:`	Axes limits.
`Legend:`	Legend for multiple curves.
`Print:`	Send to printer.

Q#01: Write a user defined MATLAB function to reconstruct $f(t)$ in the time interval of [-4, 20] using the following Fourier series; also draw Discrete Frequency Spectrum

$$f(t) = 2.5 + \frac{10}{\pi}\left(\sin\frac{\pi t}{4} + \frac{1}{3}\sin\frac{3\pi t}{4} + \frac{1}{5}\sin\frac{5\pi t}{4} + \frac{1}{7}\sin\frac{7\pi t}{4} + \dots\right)$$

Answer:

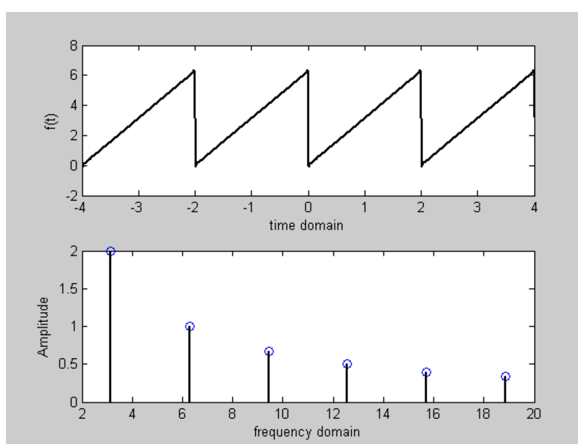
```
function [result] = four5(n)
or
function four5(n)
t = - 4:.001:20;
y = 2.5;
sum = y;
for(i =1:2:25)
sum = sum + (10/pi)*(1/i)*sin(i*pi*t/4)
end
figure(1);
plot(t,sum);
x=1 :6
w=x.*pi/4
v=((-5/pi)*1./x).*(cos(x*pi)-1)
stem(w,v)
```

Q#02: Write a user defined MATLAB function to reconstruct $f(t)$ in the time interval of [-4, 16] using the following Fourier series and also draw Discrete Frequency Spectrum

$$\mathbf{f(t)} = \pi - \sum_{n=1}^{500} \frac{2}{n}\sin n\pi t$$

Answer:

```
function [result]= four1(n)
t=-4:.001:16;
y=pi;
sum=y;
for(i=1:1:n)
sum = sum-((2/i)*sin(i*pi*t));
end
subplot(211);
```



```
plot(t,sum);
xlabel('time domain');
ylabel('f(t)');

x=1:6;
w=x.*pi;
v=2./x;%[More than one value then . (dot) is used, when single value then only / is used]
subplot(212);
stem(w,v);
xlabel('frequency domain');
ylabel('Amplitude');
```

Q#03: Write a MATLAB function to reconstruct $\boldsymbol{f(t)}$ in the time interval of [-4,16] using the following Fourier series and also draw Discrete Frequency Spectrum

$$f(t) = \frac{\pi^2}{3} + 4\sum_{n=1}^{500} \frac{(-1)^n}{n^2}\cos n\pi t$$

```
function[result]=four(n)
t=-10:0.25:10;
y=(pi*2)/3;
sum=y;
for(i=1:1:n)
   sum=sum+(4*((-1)^i)/i^2)*cos(i*pi*t);
end
   sum=sum+(4*((-1)^n)/n^2)*cos(n*pi*t);
    subplot(211);
    plot(t,sum);
    xlabel('time domain');
    ylabel('f(t)');
    x=1:10;
    w=x.*pi;
    v=(4*(-1).^x/x.*x);
    subplot(212);
    stem(w,v);
    xlabel('frequency domain');
    ylabel('amplitude');
```

Q-04:

```
a=[1 1];
b=[1 1];
c=conv(a,b)
```

Q-05:
Draw the graph of the convolution:

```
h=[1 1];
x=[1 1];
y=conv(x,h);
n=0:1:2
stem(n,y)
```

How to calculate the value of n?

Here,

h=[1 1]; means n[**0**]=1 and n[**1**]=1, that is **1** minus **o** =**1**

x=[1 1]; means n[**0**]=1 and n[**1**]=1, that is **1** minus **o** =**1**

2

Over lap will be started from 0, So 0 will be lower limit and the upper limit will be 2

Q-06:

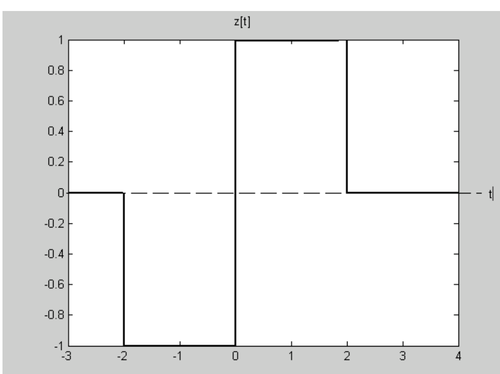
```
h=[1 2 1];
x=[2 3 -2];
y=conv(x,h);
n=0:1:4
stem(n,y)
```

Q-07:

```
t=-2:.01:4;
t0=0;
v=2*stepfun(t,t0);
plot(t,v)
```

Q-08:

Write code for the following graph;



Answer:

```
t=-2:.01:4;
t0=-2;
u=stepfun(t,t0);
v=-2*stepfun(t,t0);
t1=0;
s=2*stepfun(t,t1);
t2=2;
j=-1*stepfun(t,t2)
w=u+v+s+j;
plot(t,w)
```

Q#09:

% continuous-time shifted step function

```
t=-3:0.01:5;
t0=-2;% shifting value
u=stepfun(t,t0);
plot(t,u)
xlabel('time t')
ylabel('u(t-1)')
title('continuous-time shifted step function')
```

Q#10:

% Continuous-time rectangular pulse
```
t1=-10:-3;
t2=-3:3;
t3=3:10;

t=[t1 t2 t3];

p=[zeros(1,8) ones(1,7) zeros(1,8)];

subplot(222);

plot(t,p);
axis([-10 10 -0.5 1.5]);
grid;
title('rectangular pulse');
xlabel('time t');
ylabel('p(t)')
```

Q#11:
Write a code for the following function:
4u(t)-8u(t-1)+4u(t-2)

Answer:
```
t = - 5:.01:10;
t0 = 0;
y = 4*stepfun(t,t0);% step function means 1 at t≥0
t1=1;
z = -8*stepfun(t, t1);
t2=2;
x = 4*stepfun(t, t2);
w=y+z+x;
plot(t,w)
```

Q#12:
```
n=1:10;
step = ones(1,10);
plot(n,step);
```

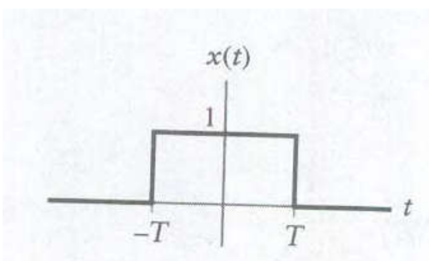
Q#13:
```
n=1:10;
step = ones(1,10);
ramp = n.*step;

subplot (211);
plot (n, step);
```

```
subplot (212);
plot(n, ramp);
```

Q#14:

```
n=1:10;

step = ones(1,10);
ramp = (n-1).*step;

subplot (211);

plot (n-1, step);
subplot (212);

plot(n-1, ramp);
```

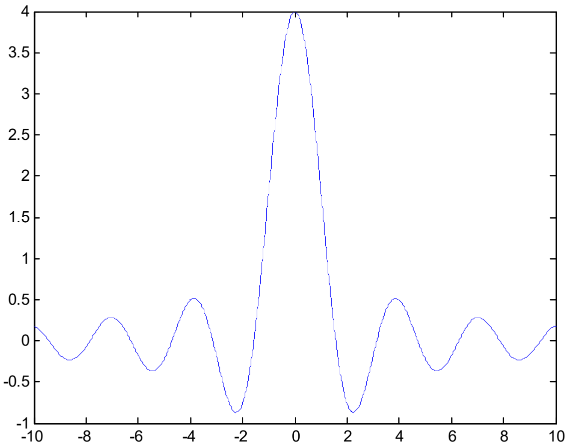
Q#15 :(Answer the question of Problem 01-Page#)



$$g(\omega) = 2T\,\frac{\sin(\omega T)}{\omega T}$$

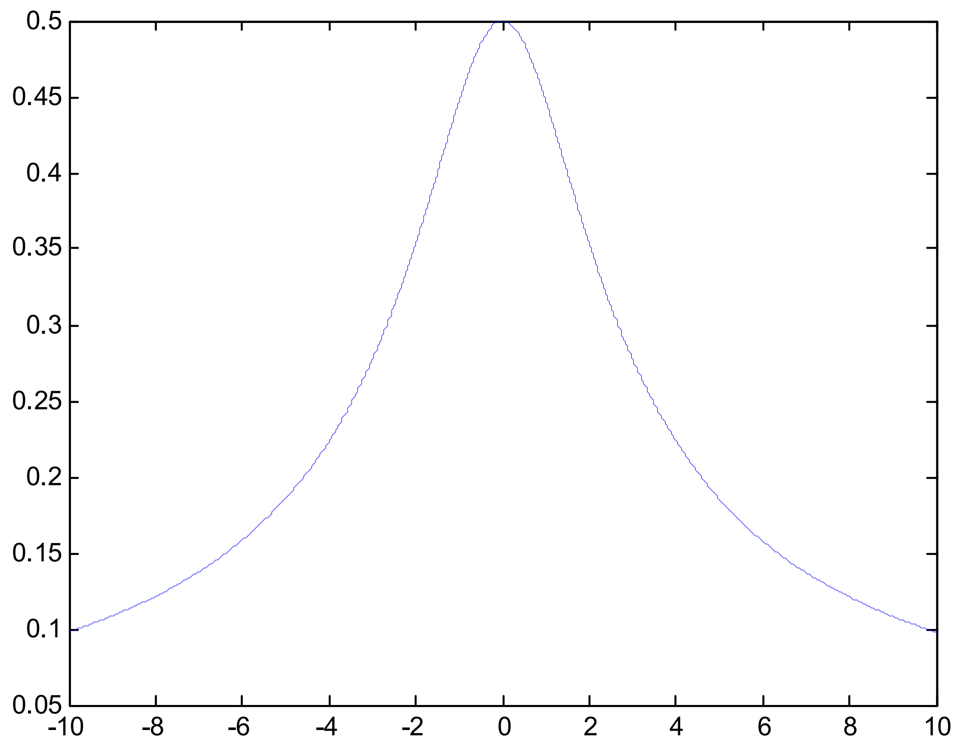
Code :

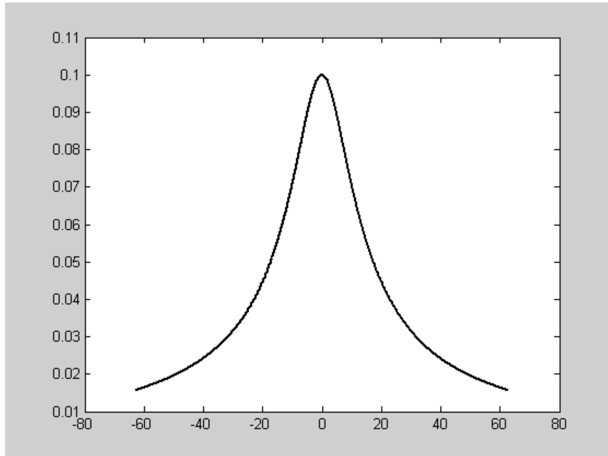
```
T = 2
w=-10:.01:10
g=2*T*(sin(T*w)./(T*w));
plot(w,g)
```



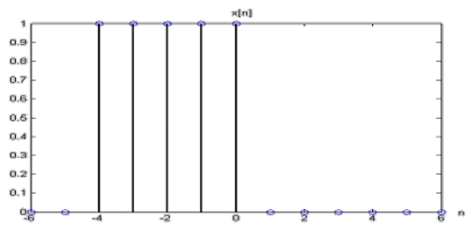
***Q#16:* (*Answer the question of Problem 02-Page#)**

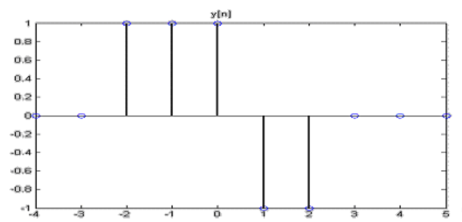
```
a=2;
w=-10:.01:10
g=1./(sqrt(a.^2+w.^2));
plot(w,g)
```





Q#17: Solution of the problem





m[n] = x[n] * y[n]

```
x=[1 1 1 1 1];
n1=-4:0;
y=[1 1 1 -1 -1];
n2=-2:2;
z=conv(x,y)
n=-6:2;
subplot(311)
stem(n1,x)
title('x')
subplot(312)
stem(n2,y)
title('y')
subplot(313)
stem(n,z)
title('z')
```

Q-18:

```
L=input('Desired length=');
n=1:L;
FT=input('Sampling frequency =');
T=1/FT;
imp =[1 zeros(1,L-1)]; %impulse function=unit amplitude at zero time and zero at any other position
```

```
step=ones(1,L);%unit amplitude at zero time axis and greater than zero
ramp=(n-1).*step; %linearly increase
subplot(311);
stem(n-1,imp);
xlabel(['Time in',num2str(T),'sec']);
ylabel('Amplitude');
subplot(312);
stem(n-1,step);
xlabel(['Time in', num2str(T),'sec']);
ylabel('Amplitude');
subplot(313);
stem(n-1,ramp);
xlabel(['Time in',num2str(T),'sec']);
ylabel('Amplitude');
```

Q-19:

```
A=1;
w0=2*pi/12;
n=-10:10;
y=A*cos(w0*n);

B=1;
r=0.85;
x=B*r.^n;%decaying exponential
z=x.*y;%elementwise multiplication
figure(1)
subplot(311)
stem(n,y)
subplot(312)
figure(2)
stem(n,x)
subplot(313)
figure(3)
stem(n,z)
```

Q-20:

```
%Generation of real exponential sequence
a=input('Type in exponent=');
k=input('type in the gain constant=');
N=input('type in length of sequence=');
n=0:N;
x=k*a.^n;
```

```
stem(n,x);
xlabel('Time index n');
ylabel('Amplitude');
title(['\alpha=',num2str(a)]);
clear;                          % clear all variables
clf;                           % clear all figures
```

% Compute yce, the Fourier Series in complex exponential form

```
yce = c0*ones(size(t));           % initialize yce to c0

for n = -N:2:N,                 % loop over series index n
cn = 2/(j*n*wo);                % Fourier Series Coefficient
yce = yce + cn*exp(j*n*wo*t);    % Fourier Series computation
end

subplot(2,1,1)
plot([-3 -2 -2 -1 -1  0 0 1  1  2 2 3],...    % plot original y(t)
[-1 -1  1  1 -1 -1 1 1 -1 -1 1 1], ':');
hold;
plot(t,yce);
grid;
xlabel('t (seconds)')
ylabel('y(t)');
ttle = ['EE341.01: Truncated Exponential Fourier Series with N = ',...
        num2str(N)];
title(ttle);
hold
```

% Compute yt, the Fourier Series in trigonometric form

```
yt = c0*ones(size(t));           % initialize yce to c0
for n = 1:2:N,                  % loop over series index n
cn = 2/(j*n*wo);                % Fourier Series Coefficient
 yt = yt + 2*abs(cn)*cos(n*wo*t+angle(cn));  % Fourier Series computation
end
subplot(2,1,2)
plot([-3 -2 -2 -1 -1  0 0 1  1  2 2 3],...    % plot original y(t)
[-1 -1  1  1 -1 -1 1 1 -1 -1 1 1], ':');
hold;
plot(t,yt);
grid;
xlabel('t (seconds)')
ylabel('y(t)');
ttle = ['EE341.01: Truncated Trigonometric Fourier Series with N = ',...
        num2str(N)];
title(ttle);
hold
```

% Draw the amplitude spectrum from exponential Fourier Series

```
figure(2)                       % put next plots on figure 2
subplot(2,1,1)
stem(0,c0);                      % plot c0 at nwo = 0
hold
for n = -N:2:N,                  % loop over series index n
cn = 2/(j*n*wo);                 % Fourier Series Coefficient
stem(n*wo,abs(cn))                 % plot |cn| vs nwo
end
for n = -N+1:2:N-1,                % loop over even series index n
 cn = 0;                       % Fourier Series Coefficient
 stem(n*wo,abs(cn)*180/pi);       % plot |cn| vs nwo
end
xlabel('w (rad/s)')
ylabel('|cn|')
ttle = ['EE341.01: Amplitude Spectrum with N = ',num2str(N)];
title(ttle);
hold
```

% Draw the phase spectrum from exponential Fourier Series

```
subplot(2,1,2)
stem(0,angle(c0)*180/pi);                 % plot angle of c0 at nwo = 0

hold
for n = -N:2:N,                  % loop over odd series index n
 cn = 2/(j*n*wo);                 % Fourier Series Coefficient
 stem(n*wo,angle(cn)*180/pi);     % plot |cn| vs nwo
end
for n = -N+1:2:N-1,                % loop over even series index n
 cn = 0;                       % Fourier Series Coefficient
stem(n*wo,angle(cn)*180/pi);     % plot |cn| vs nwo
end

xlabel('w (rad/s)')
ylabel('angle(cn) (degrees)')
ttle = ['EE341.01: Phase Spectrum with N = ',num2str(N)];
title(ttle);
hold
```

MATLAB Plots Generated:

```
Function Square (t, n);
{Sum of n sine waves, n: number of sine waves, t: time}
Begin
f := 1k;
 w := 2 * pi * f;
```

```
 x := 0;
For i := 0 To n - 1 Do
 x := x + 1 / (2 * i + 1) * sin((2 * i + 1) * w * t);
Square := 4 / pi * x;
End;
{Sum of 20 sine waves}
Draw(Square(Time, 20), Square);
```

Q-12: Draw the following complex wave using MATLAB

$$f(t) = \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{1}{2n-1} \cos nt + \sum_{n=1}^{\infty} \frac{(-1)^n}{2n} \sin nt$$

```
function [result] = faisal(n)
t = - 4:.01:16;
y = pi/2;
sum = y;
for(i =1:1:n)
sum = sum + (1./(2*i-1))*cos(i*t)+((-1.^i)./(2*i))*sin(i*t)
end
plot(t,sum);
```

Q-13: Draw the line spectrum for the signal

$$f(t) = \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{1}{2n-1} \cos nt + \sum_{n=1}^{\infty} \frac{(-1)^n}{2n} \sin nt$$

```
n= 1:6
w=n
a=1./(2*n-1)
b=(-1.^n)./(2*n)
r=sqrt(a.^2+b.^2)
stem(w,r)
```

Q-14: Draw the following complex wave using MATLAB

$$f(t) = \pi - \sum_{n=1}^{500} \frac{2}{n} \sin n\pi t$$

```
function [result]= four1(n)
t=-4:.01:4;
y=pi;;
sum=y;
for(i=1:1:n)
sum = sum-((2/i)*sin(i*pi*t))
end
```

```
plot(t,sum);
xlabel('time domain');
ylabel('f(t)');
```

Q-15: Draw the line spectrum for the signal

$$f(t) = \pi - \sum_{n=1}^{500} \frac{2}{n} \sin n\pi t$$

```
n= 1:6
w=n*pi
a=0
b=-2./n
r=sqrt(a.^2+b.^2)
stem(w,r)
```

Q-5: Let, $x = \frac{1}{4}\cos 8t + \frac{1}{8}\sin 8t$

What is the amplitude of the resultant signal?

```
A1=1/4;
B1=1/8;
t=0:0.01:1;
w=8;
y=A1*cos(w*t);
z=B1*sin(w*t);
x=y+z
plot(t,y,t,z,t,x)
```

Q-6: Draw the sine wave and cosine wave with Phase difference is 90^0

```
t=-0.5:0.01:1;
A=1;
B=1;
f=1;
w=2*pi*f;
b=0
c=pi/2
y=A*sin(w*t+b);
z=B*sin(w*t+c);
plot(t,y,t,z)
```

Convolution

```
x=[1 1 ];
n1=0:1;
y=[1 1 ];
n2=0:1;
```

```
z=conv(x,y)
n=0:2;
subplot(311)
stem(n1,x)
title('x')
subplot(312)
stem(n2,y)
title('y')
subplot(313)
stem(n,z)
title('z')
```